

Project Design Phase-II

Data Flow Diagram & User Stories

Date	30 September 2026
Team ID	SWTID-2026-1542
Project Name	AnswerBot – AI-powered FAQ and customer support automation
Maximum Marks	4 Marks

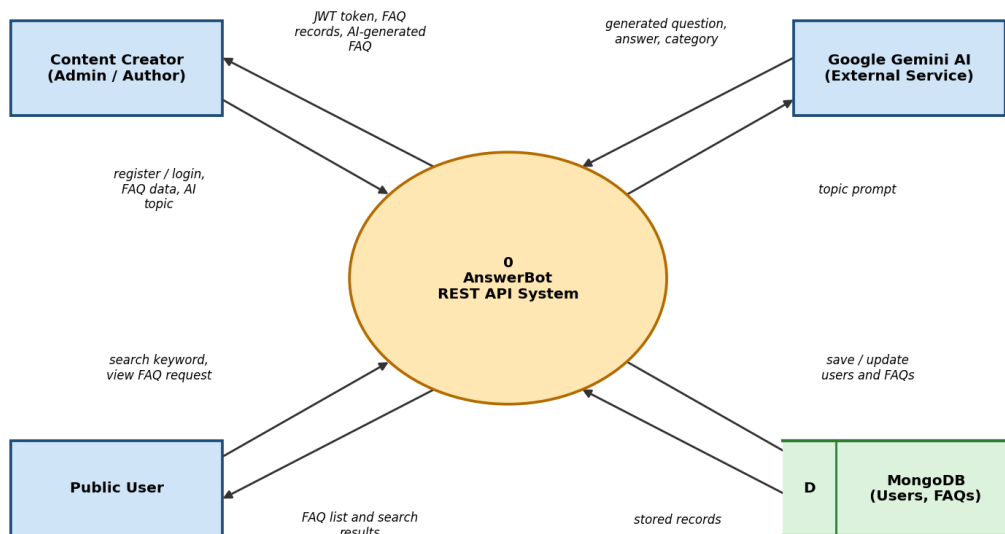
Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Level 0 (Context Diagram) – AnswerBot

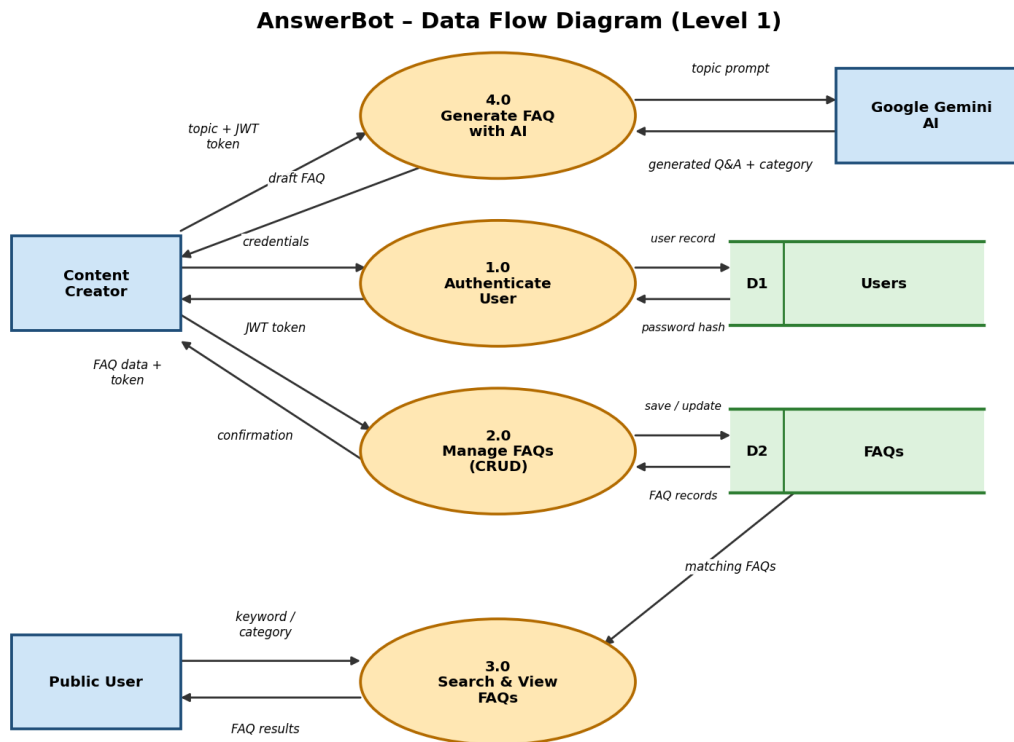
At the highest level, AnswerBot is one system that exchanges data with content creators, public users, the Google Gemini AI service and the MongoDB database.

AnswerBot - Data Flow Diagram (Level 0 / Context Diagram)



DFD Level 1 – AnswerBot

Level 1 breaks the system into four processes and two data stores.



Process	Name	Data flow	Data store / service
1.0	Authenticate User	Content Creator → credentials; returns JWT token	D1 Users (bcrypt password hash)
2.0	Manage FAQs (CRUD)	Content Creator → FAQ data + token; returns confirmation	D2 FAQs (createdBy reference)
3.0	Search & View FAQs	Public User → keyword / category; returns FAQ results	D2 FAQs (read only)
4.0	Generate FAQ with AI	Content Creator → topic; Gemini AI → generated Q&A + category; returns draft FAQ	External: Google Gemini AI

User Stories

The table below lists the user stories for AnswerBot. User story numbers match the product backlog in the Project Planning Phase.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Content Creator	Registration	USN-2	As a user, I can register by entering my name, email and password.	I can create an account and my password is stored encrypted	High	Sprint-1
Content Creator	Login	USN-4	As a user, I can log in with my email and password.	I receive a JWT token and can access protected endpoints	High	Sprint-1
Content Creator	FAQ Management	USN-8	As a creator, I can create an FAQ with question, answer and category.	FAQ is saved in MongoDB with my user as creator	High	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Content Creator	FAQ Management	USN-11	As a creator, I can update an FAQ that I created.	Changes are saved and updatedAt changes	High	Sprint-2
Content Creator	FAQ Management	USN-12	As a creator, I can delete an FAQ that I created.	FAQ is removed; other users' FAQs cannot be deleted by me	High	Sprint-2
Content Creator	AI FAQ Generation	USN-17	As a creator, I can generate a question-answer pair from a topic using AI.	Gemini returns a question, answer and category I can save	High	Sprint-3
Public User	FAQ Management	USN-9	As a user, I can view the list of published FAQs without logging in.	FAQ list with count is returned with no token	High	Sprint-2
Public User	Category Management	USN-15	As a user, I can filter FAQs by category.	Only FAQs of the chosen category are shown	Medium	Sprint-2
Public User	Search	USN-19	As a user, I can search FAQs by keyword.	Matching FAQs are returned for the keyword	High	Sprint-3
Administrator	FAQ Management	USN-26	As an administrator, I can view, update or delete any FAQ.	Admin can change or remove any FAQ record	Medium	Future release
Administrator	User Management	USN-27	As an administrator, I can manage registered users and FAQ categories.	Admin can view users and add/edit categories	Medium	Future release